USN | | | | | | | | | | | [22CSE156]

FIFTH SEMESTER B. Tech EXAMINATION, DECEMBER - 2024

ESSENTIALS OF MACHINE LEARNING

Time: 3 Hours **Maximum Marks: 100**

Instructions: **I.** Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A			**5 x2=10**
1.	**a.**	Discuss machine learning with its importance.	L2	CO1	**02**
	b.	Differentiate linear and logistic regression.	L3	CO2	**02**
	c.	Illustrate the concept of Principal Component Analysis (PCA).	L2	CO3	**02**
	d.	Interpret the role of entropy in decision tree learning.	L2	CO4	**02**
	e.	Outline the K-Means clustering and its main objective.	L2	CO5	**02**
		PART-B			**6 X 5=30**
2.	**a.**	Explain the different types of learning in machine learning.	L2	CO1	**05**
		OR			
	b.	Summarize the role of exploratory data analysis (EDA) in machine learning.	L2	CO1	**05**
3.	**a.**	Describe how Ordinary Least Squares (OLS) is used in linear regression?	L2	CO2	**05**
		OR			
	b.	Compare logistic regression with Naive Bayes classification.	L2	CO2	**05**
4.	**a.**	Illustrate the different types of kernels used in support vector machines (SVM).	L2	CO2	**05**
		OR			
	b.	Sketch the concept of confusion matrix and its significance in classification models.	L3	CO2	**05**
5.	**a.**	Differentiate the hyperparameter tuning and its importance in model optimization.	L4	CO3	**05**
		OR			
	b.	Contrast the Principal Component Analysis (PCA) and the Linear Discriminant Analysis (LDA).	L4	CO3	**05**
6.	**a.**	Interpret the ID-3 Algorithm in decision tree learning and its key steps.	L2	CO4	**05**
		OR			
	b.	Illustrate the key issues in decision tree learning.	L2	CO4	**05**

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7.	a.	How does hierarchical clustering differ from K-Means clustering?	L3	CO5	05
		OR			
	b.	Demonstrate the difference between reinforcement learning and supervised learning?	L2	CO6	05
		PART-C		**6 X 10=60**	
8.	a.	Describe the process of designing a machine learning system with an example.	L2	CO1	10
		OR			
	b.	Discuss the role of NumPy and Pandas in data preprocessing and analysis for machine learning.	L2	CO1	10
9.	a.	Employ the working of logistic regression and how it can be used for binary classification.	L3	CO2	10
		OR			
	b.	Illustrate the performance of a linear machine learning model using confusion matrix, accuracy, precision, and recall.	L3	CO2	10
10.	a.	Discuss the role of hyperplane and decision surface in support vector machines (SVM).	L2	CO2	10
		OR			
	b.	Explain the concept of Maximum Likelihood Estimation (MLE) and its application in parameter estimation.	L2	CO3	10
11.	a.	Demonstrate how Principal Component Analysis (PCA) is used for dimension reduction and how it improves model performance.	L3	CO3	10
		OR			
	b.	Dramatize the entropy and information gain concepts in decision tree learning with an example.	L3	CO4	10
12.	a.	Differentiate the concept of Random Forest and how it enhances decision tree learning.	L4	CO4	10
		OR			
	b.	Compare and contrast K-Means clustering and DBSCAN in terms of algorithm and real-world applications.	L4	CO5	10
13.	a.	Dramatize the reinforcement learning and describe the methods used to teach a system to maximize rewards.	L3	CO6	10
		OR			
	b.	Demonstrate how does fixing the value of K affects the performance of K-Means clustering? Explain with examples.	L3	CO5	10